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23 March 1972
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MILITARY SPECIFICATION

FLIGHT TESTING, ELECTRIC SYSTEM, PILOTED AIRCRAFT AND GUIDED MISSILE, GENERAL REQUIREMENTS

1. SCOPE

1.1 Scope. This specification covers the general requirements for flight testing of electric systems of piloted aircraft and guided missiles.

1.2 Objective. The objective of this specification is to require sufficient flight tests to insure that a satisfactory electric system is installed in the aircraft or guided missile.

2. APPLICABLE DOCUMENTS

* 2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

STANDARDS

Military

MIL-STD-831 Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 General requirements. Prior to, and throughout the test program, the equipment shall be inspected for loose and broken parts, chafed insulation, and general condition. Brush covers, blast-air housing, or other cover plates shall be removed during inspection. On rotating machines, the condition of the slip rings, commutator, brushes, et cetera, shall be noted. Upon completion of the inspection, all parts shall be reassembled in their proper condition. Lock wiring shall be replaced where necessary.

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3.2 Materials

3.2.1 Discrepancies. Any discrepancies in material or equipment furnished by the Government (GFAE) or deviations from its detail specification, or evidence of poor workmanship shall be brought to the attention of the procuring activity as soon as practicable.

3.3 Instrumentation. All instruments shall be calibrated together with their leads, current transformers, or shunts prior to the start of testing. The overall error of measurement shall not exceed ±5 percent on aircraft-type instrumentation and ±3 percent on laboratory instrumentation (oscillograph excepted).

3.4 Load banks. Load banks may be used to simulate the power requirements of actual load equipment provided the substitution will not adversely affect the system test results.

* 3.5 System tests. The electric system (see 6.3.1) shall be flight tested as specified herein. These tests may be performed concurrently with flight of the prototype aircraft or guided missile. Flight tests shall cover all designed flight conditions. Particular attention shall be given to expected marginal areas such as reduced generating capacity due to operation at high altitudes, with minimum cooling, and at low engine speeds incident to approach, letdown, and taxi.

3.5.1 Electrical tests. Sufficient flight tests shall be performed to assure an electric system that is satisfactory in all respects and particularly so from the standpoint of performance, safety, and reliability. These tests shall:

a. Take into account all possible normal operating conditions. Emergency and abnormal conditions such as faults, component failures, malfunctioning of equipment, and improper operating procedures shall be taken into account where applicable.

b. Demonstrate that the system fulfills its design requirements and objectives. This shall entail the functioning of all system components installed in the aircraft or guided missile.

c. Establish the allowable adjustment tolerances of control elements based on system performance.

d. Determine that the system electric power characteristics conform to the applicable military specification.

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3.5.1.1 Electrical test items. The system shall be tested to determine the system characteristics or the effect on the system of those items listed below, where applicable. If necessary to assure compliance with the requirements of this specification, or at the option of the testing facility, the list shall be supplemented with additional items.

a. Voltage

(1) Regulation - main bus or regulation point (steady state)

(2) Regulation - load bus or load (steady state)

(3) Phase unbalance

(4) Transient limits and stability

(5) Phase displacement

(6) Waveform (crest factor and harmonic content)

(7) Amplitude modulation

(8) Phase rotation

(9) Ripple

(10) Interruption (power transfer or fault clearing)

(11) Drift (during warmup)

b. Frequency

(1) Regulation (steady state)

(2) Transient limits and stability

(3) Modulation

c. Capacity

(1) Normal

(2) Overload

(3) Fault

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(4) Minimum

d. Parallel operation

(1) Real-load division

(2) Reactive-load division

(3) Stability

(4) Synchronizing

e. Manual controls

(1) Range of adjustments

f. Protection

(1) Undervoltage

(2) Overvoltage

(3) Underfrequency

(4) Overfrequency

(5) Ceiling excitation

(6) Generator, wire, and bus fault (including interrupting capacity and coordination)

(7) Reverse current

(8) Reverse power

(9) Miscellaneous control-system faults

(10) Coordination

g. Loads

(1) Switching

(2) Automatic transfer

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(3) Voltage self-modulation

(4) Compatibility with power supply.

3.5.2 Environmental tests. Sufficient flight tests shall be conducted on the electric system to uncover any adverse effects resulting from the operational environments of the aircraft or missile.

3.6 Dimensions. Pertinent outside dimensions, weight, and terminal marking and location shall be noted for major items of electrical equipment and compared with the applicable specification requirements.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Test procedure. When specified in the contract (see 6.2), a detailed test procedure applicable to the particular aircraft or guided missile shall be prepared by the contractor and submitted for approval of the procuring activity prior to actual testing. The test procedure shall cover all requirements of this specification and shall include the following:

a. System description

b. Purpose of each test

c. Detailed procedure for each test

d. Required instrumentation for each test

e. Type, manufacturer, and function of test components, noting whether CFAE or GFAE.

4.2 Flight tests. Testing shall be done in accordance with the requirements of this specification. Tests performed in accordance with the approved test procedure shall be considered to fulfill the test requirements of this specification. Minor details of the approved test procedure may be modified without procuring activity approval provided the purpose of the applicable test is not changed.

* 4.3 Test report. When specified in the contract (see 6.2), the contractor shall submit a final report of the flight testing of the aircraft or guided missile electric systems and equipment. This report should follow, as nearly as practicable, the logical sequence and the headings and subheadings of the tests outlined in the contractor's test procedure as approved by the procuring activity. The report shall be prepared in accordance with MIL-STD-831 and shall include the following:

a. Detailed system description

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b. Exact testing procedure followed and reasons for any changes from the approved test procedure

c. Detailed results of the testing

d. Serial, type, and manufacturer of test instruments

e. Serial, type, and manufacturer of test components, including applicable specifications

f. Applicable environmental test conditions

g. Appropriate photographs, sketches, diagrams, and curves to explain test installations and results

h. Recommendations and conclusions

i. An appendix recording all pertinent changes made in the system (including circuitry and equipment) during testing, and reasons for each change.

5. PREPARATION FOR DELIVERY

* This section is not applicable to this specification.

6. NOTES

6.1 Intended use. This specification is intended for use by contractors in conducting flight tests of aircraft or guided missile electric systems.

* 6.2 Data. Data generated by this specification is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military department's Authorized Data List (ADL). The data produced by this specification is as follows:

a. Test procedure (see 4.1)

b. Test report (see 4.3).

6.3 Definition

6.3.1 Electric system. The term electric system, as used herein, should be construed to mean power-generation equipment, batteries, conversion equipment, distribution circuits, and all associated control and protection equipment. Load equipment should be considered only to the extent of its effect on, or the extent to which it is affected by the operation of the system.

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6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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